

BEE 4750 Homework 2: Systems Modeling and Simulation

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Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

Activating project at `c:\Users\adric\OneDrive - Cornell University\Documents\3. Env Sys A

Installed Accessors	v0.1.42
Installed JpegTurbo_jll	v3.1.3+0
Installed SentinelArrays	v1.4.8
Installed WorkerUtilities	v1.6.1
Installed InlineStrings	v1.4.5
Installed InvertedIndices	v1.3.1
Installed Roots	v2.2.10
Installed DataFrames	v1.8.0
Installed ConstructionBase	v1.6.0
Installed DataValueInterfaces	v1.0.0
Installed WeakRefStrings	v1.4.2
Installed TableTraits	v1.0.1
Installed PooledArrays	v1.4.3
Installed Ghostscript_jll	v9.55.1+0
Installed CommonSolve	v0.2.4
Installed CompositionsBase	v0.1.2
Installed InverseFunctions	v0.1.17
Installed Crayons	v4.1.1
Installed Latexify	v0.16.10
Installed PrettyTables	v3.0.8
Installed Glib_jll	v2.86.0+0
Installed Tables	v1.12.1
Installed FilePathsBase	v0.9.24
Installed IteratorInterfaceExtensions	v1.0.0
Installed StringManipulation	v0.4.1
Installed CSV	v0.10.15

Precompiling project...

19051.0 ms	IteratorInterfaceExtensions
13421.4 ms	InverseFunctions
13447.2 ms	WorkerUtilities
13413.0 ms	CommonSolve
13400.0 ms	DataValueInterfaces
13377.3 ms	InlineStrings
13357.1 ms	InvertedIndices
13348.6 ms	ConstructionBase
13372.9 ms	CompositionsBase
13324.4 ms	PooledArrays
13339.2 ms	Crayons
910.1 ms	TableTraits
1126.2 ms	LogExpFunctions → LogExpFunctionsInverseFunctionsExt
1145.2 ms	InverseFunctions → InverseFunctionsDatesExt
1221.5 ms	InlineStrings → ParsersExt

1665.5 ms	JpegTurbo_jll
1888.6 ms	InverseFunctions → InverseFunctionsTestExt
1202.7 ms	ConstructionBase → ConstructionBaseLinearAlgebraExt
1920.8 ms	Unitful → InverseFunctionsUnitfulExt
2349.9 ms	FilePathsBase
1080.4 ms	CompositionsBase → CompositionsBaseInverseFunctionsExt
2685.1 ms	SentinelArrays
2787.1 ms	Glib_jll
2001.1 ms	Unitful → ConstructionBaseUnitfulExt
1910.0 ms	Tables
1838.7 ms	WeakRefStrings
3759.8 ms	StringManipulation
1396.0 ms	FilePathsBase → FilePathsBaseMmapExt
2006.6 ms	Libtiff_jll
2397.5 ms	Ghostscript_jll
2542.5 ms	FilePathsBase → FilePathsBaseTestExt
2312.0 ms	Cairo_jll
3911.1 ms	Accessors
4789.1 ms	Qt6Base_jll
1771.1 ms	Accessors → TestExt
1782.8 ms	Accessors → LinearAlgebraExt
1919.8 ms	Accessors → UnitfulExt
2723.1 ms	HarfBuzz_jll
4945.6 ms	Latexify
2315.3 ms	libass_jll
2186.3 ms	Pango_jll
2341.6 ms	UnitfulLatexify
2762.6 ms	Latexify → SparseArraysExt
4945.8 ms	Roots
1529.4 ms	Roots → RootsUnitfulExt
3859.1 ms	FFMPEG_jll
7467.8 ms	Qt6ShaderTools_jll
2679.0 ms	FFMPEG
3212.8 ms	GR_jll
3707.9 ms	Qt6Declarative_jll
19783.7 ms	CSV
5813.9 ms	GR
33530.6 ms	PrettyTables
49376.4 ms	DataFrames
7305.2 ms	Latexify → DataFramesExt
79491.0 ms	Plots
5267.1 ms	Plots → UnitfulExt

57 dependencies successfully precompiled in 150 seconds. 162 already precompiled.

```
Pkg.add("GraphRecipes")
```

```
Updating registry at `C:\Users\adric\.julia\registries\General.toml`  
Resolving package versions...
```

```
Installed StaticArrays    v1.9.15  
Installed SimpleTraits    v0.9.5  
Installed Pango_jll       v1.56.4+0  
Installed PrettyTables    v3.0.10  
Installed Interpolations  v0.16.2  
Installed ChainRulesCore  v1.26.0  
Installed Graphs          v1.13.1  
Installed ColorSchemes    v3.31.0
```

```
Updating `C:\Users\adric\OneDrive - Cornell University\Documents\3. Env Sys Analysis\hw2`  
[bd48cda9] + GraphRecipes v0.5.14
```

```
Updating `C:\Users\adric\OneDrive - Cornell University\Documents\3. Env Sys Analysis\hw2`  
[1520ce14] + AbstractTrees v0.4.5  
[79e6a3ab] + Adapt v4.3.0  
[ec485272] + ArnoldiMethod v0.4.0  
[13072b0f] + AxisAlgorithms v1.1.0  
[d360d2e6] + ChainRulesCore v1.26.0  
[35d6a980] ↑ ColorSchemes v3.30.0 v3.31.0  
[3da002f7] ↓ ColorTypes v0.12.1 v0.11.5  
[c3611d14] ↓ ColorVectorSpace v0.11.0 v0.10.0  
[411431e0] + Extents v0.1.6  
[5c1252a2] + GeometryBasics v0.5.10  
[4d00f742] + GeometryTypes v0.8.5  
[bd48cda9] + GraphRecipes v0.5.14  
[86223c79] + Graphs v1.13.1  
[d25df0c9] + Inflate v0.1.5  
[a98d9a8b] + Interpolations v0.16.2  
[c8e1da08] + IterTools v1.10.0  
[46757867] + NetworkLayout v0.4.10  
[6fe1bfb0] + OffsetArrays v1.17.0  
[08abe8d2] ↑ PrettyTables v3.0.8 v3.0.10  
[c84ed2f1] + Ratios v0.4.5  
[699a6c99] + SimpleTraits v0.9.5  
[90137ffa] + StaticArrays v1.9.15  
[1e83bf80] + StaticArraysCore v1.4.3  
[efce3f68] + WoodburyMatrices v1.0.0  
[5ae413db] + EarCut_jll v2.2.4+0  
[36c8627f] ↑ Pango_jll v1.56.3+0 v1.56.4+0  
[8ba89e20] + Distributed v1.11.0
```

```

[1a1011a3] + SharedArrays v1.11.0
      Info Packages marked with  and  have new versions available. Those with  may be upg
Precompiling project...
 7729.8 ms   SimpleTraits
 6825.3 ms   EarCut_jll
 7329.0 ms   ChainRulesCore
 8047.2 ms   Pango_jll
 1943.9 ms   ChainRulesCore → ChainRulesCoreSparseArraysExt
 1286.1 ms   Roots → RootsChainRulesCoreExt
 2208.5 ms   LogExpFunctions → LogExpFunctionsChainRulesCoreExt
11037.8 ms   ColorSchemes
15030.1 ms   StaticArrays
 1484.8 ms   Adapt → AdaptStaticArraysExt
 1512.1 ms   StaticArrays → StaticArraysStatisticsExt
 1597.5 ms   ConstructionBase → ConstructionBaseStaticArraysExt
 1808.7 ms   StaticArrays → StaticArraysChainRulesCoreExt
 1756.3 ms   Accessors → StaticArraysExt
 2496.6 ms   ArnoldiMethod
 3217.8 ms   GeometryTypes
 3307.0 ms   Interpolations
 2072.3 ms   Interpolations → InterpolationsUnitfulExt
 6203.7 ms   Graphs
14996.3 ms   GeometryBasics
 3305.2 ms   NetworkLayout
 3430.1 ms   NetworkLayout → NetworkLayoutGraphsExt
37834.2 ms   PrettyTables
16788.3 ms   GraphRecipes
111536.6 ms  Plots → GeometryBasicsExt
25 dependencies successfully precompiled in 157 seconds. 230 already precompiled.
2 dependencies precompiled but different versions are currently loaded. Restart julia to a

```

```

using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
using GraphRecipes

```

```

Warning: Module Plots with build ID fafbfcfd-7caf-33b3-0000-512bf3d02c4d is missing from th
This may mean Plots [91a5bcdd-55d7-5caf-9e0b-520d859cae80] does not support precompilation
@ Base loading.jl:2541

```

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO₂ concentrations, and ocean CO₂ concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

Tip

Think about Henry's law for CO₂.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

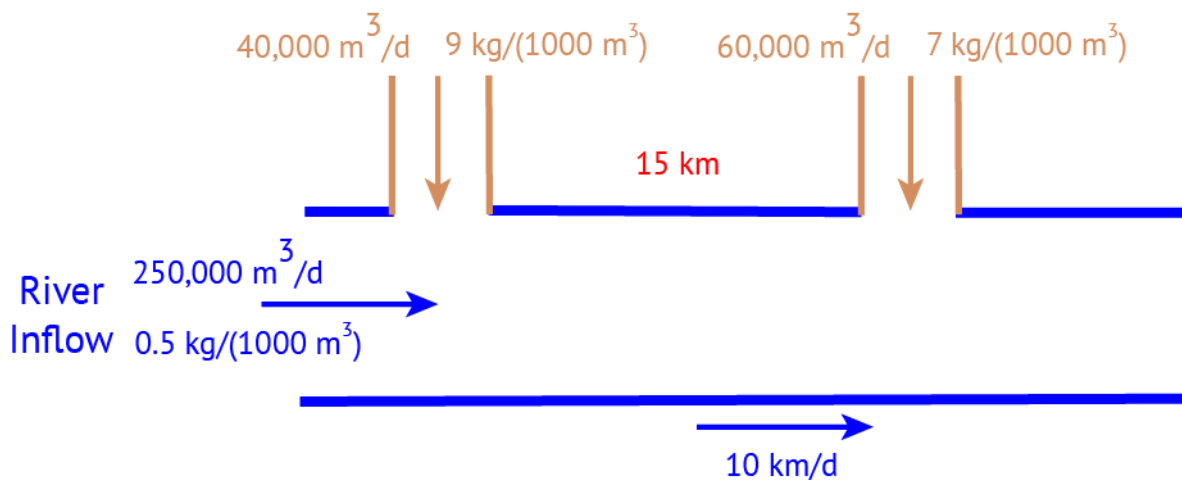


Figure 1: river_diagram.png

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/out-flows between these boxes. How did you decide how many boxes were needed?

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

Tip

Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

Problem 2.3

Determine if the system is in compliance with a regulatory limit of 2.3 kg/(1000m³). You can do this analytically or computationally.

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized “Snowball Earth”. In this problem, we’ll introduce a simple model of the Earth’s energy balance which includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C \frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1 - \alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A - BT)}_{\text{outgoing radiation}} + \underbrace{a \ln \left(\frac{[CO_2]}{[CO_2]_{PI}} \right)}_{\text{greenhouse effect}}, \quad (1)$$

where:

- T is the Earth’s global mean temperature (in °C);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be 51 J/m²/°C;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;
- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m² but during the Neoproterozoic Era had a value of 1272 W/m² (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);
- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3$ W/m²/°C (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2$ W/m² (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1$ yr.

Problem 3.2

Rather than assuming a constant value for the albedo α , we will represent the ice-albedo feedback by letting α depend on T :

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^\circ\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T + 10)/20) & \text{if } -10^\circ\text{C} \leq T \leq 10^\circ\text{C} \\ \alpha_0 & \text{if } T \geq 10^\circ\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoproterozoic value of S with temperature-varying albedo for initial values of T spanning $-60^\circ\text{C} \leq T_0 \leq 30^\circ\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C ?

References

List any external references consulted, including classmates.